

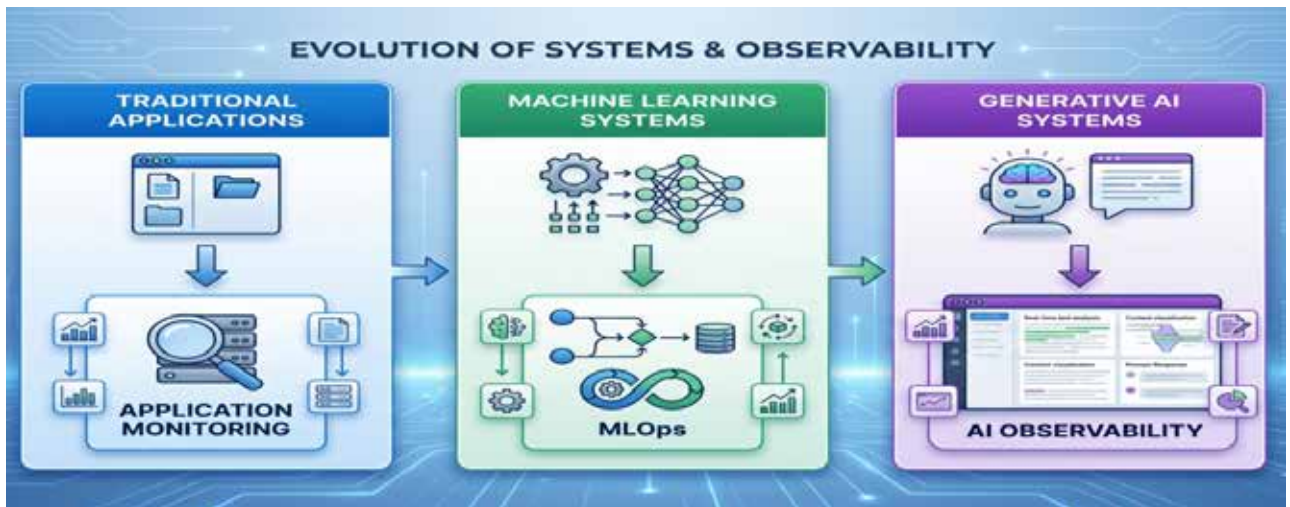


Figure 2: Evolution of operational practices from application monitoring to AI observability

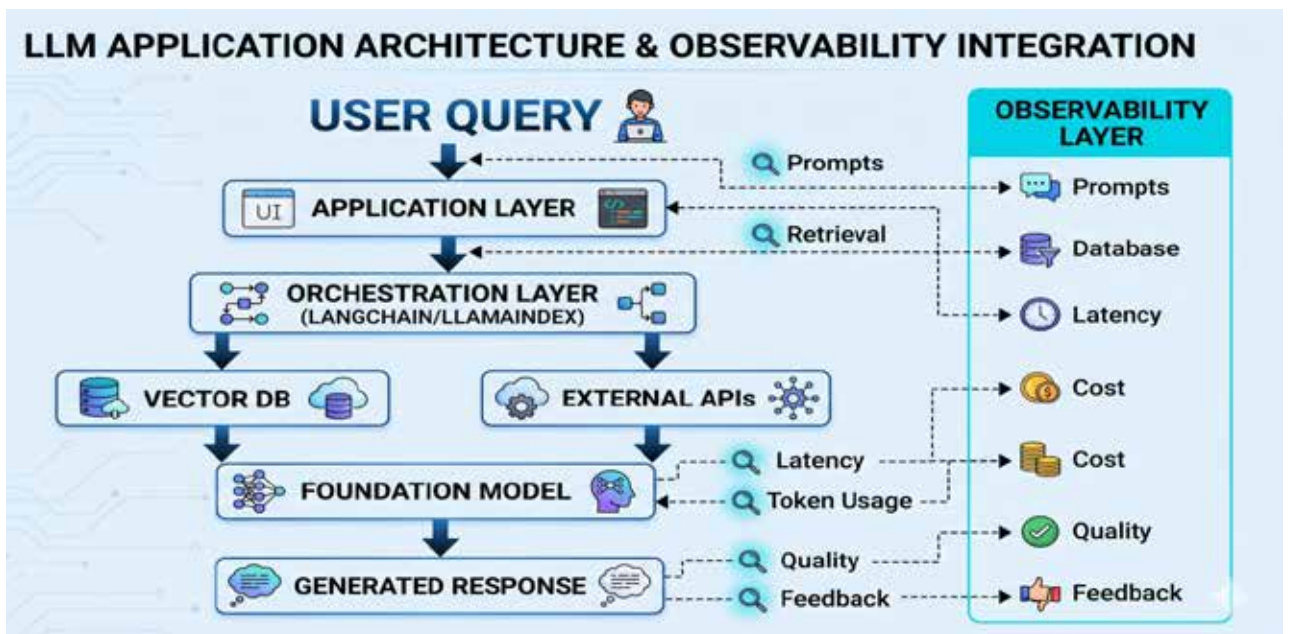


Figure 3: Building blocks of a modern AI application workflow

prompts, retrieval workflows, model behaviour, response quality, latency, token consumption, and user feedback. This transition signals a new operational paradigm where monitoring expands from infrastructure to the behaviour and efficacy of AI systems themselves.

Why observability matters for AI workloads and LLM applications

The models themselves are becoming

less important as organisations deploy more AI-powered systems in production environments. While model development often gets a lot of attention, the long-term success of an AI application depends on how well it can be monitored, maintained, and improved after deployment.

AI systems consist of multiple interconnected parts that collaborate to generate an answer, unlike conventional software applications. These can include

foundation models, embedding services, vector databases, retrieval pipelines, orchestration frameworks, guardrails, and external APIs. A broken or degraded element directly affects the quality of the final output.

The importance of observability increases when AI applications start to serve real users.

For example, consider an enterprise knowledge assistant built on RAG. A user can ask a perfectly